

Advanced (Habanero)

- Q1.** What is the unit rate of 12 miles in 4 hours?
(A) 1 mile/hour (A) rac14
(B) 2 miles/hour rac13
(C) 3 miles/hour (A)
(D) 4 miles/hour rac12
(E) 6 miles/hour (A) rac23
- Q2.** A car travels 250 miles in 5 hours. What is its speed in miles per minute? rac34
(A) 1 mile/minute
(B) 5 miles/minute
(C) 8 miles/minute
(D) 10 miles/minute
(E) 50 miles/minute
- Q3.** If y varies directly with x , and $y = 12$ when $x = 4$, what is the equation of the variation?
(A) $y = 2x$
(B) $y = 3x$
(C) $y = 4x$
(D) $y = 5x$
(E) $y = 6x$
- Q4.** If y varies inversely with x , and $y = 6$ when $x = 2$, what is y when $x = 3$?
(A) 3
(B) 4
(C) 6
(D) 8
(E) 12
- Q5.** A worker can complete a task in 8 hours. What fraction of the task can the worker complete in 2 hours?
(A)
- Q6.** A water tank can hold 1200 gallons of water. If 300 gallons of water are already in the tank, what percentage of the tank is filled?
(A) 20
(A) 25
(A) 30
(A) 40
(A) 50
- Q7.** A car travels from city A to city B at an average speed of 60 miles per hour and returns at an average speed of 40 miles per hour. What is the average speed for the round trip?
(A) 40 miles/hour
(B) 48 miles/hour
(C) 50 miles/hour
(D) 55 miles/hour
(E) 60 miles/hour
- Q8.** What is the unit rate of 480 yards in 8 minutes?
(A) 50 yards/minute
(B) 60 yards/minute
(C) 70 yards/minute
(D) 80 yards/minute
(E) 90 yards/minute

Open Ended Questions (FRQ)

- Q9.** A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. If the ratio of whole wheat to white bread is 3 : 5, how many loaves of whole wheat bread are sold per day? Show your work and explain your reasoning.
- Q10.** A water tank is being filled at a rate of 5 gallons per minute. If the tank can hold 1200 gallons of water, how many minutes will it take to fill the tank? If the water is then drained from the tank at a rate of 2 gallons per minute, how many minutes will it take to drain the tank? Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully and choose the correct answer.
- Check your work and make sure you have shown all necessary calculations.
- Manage your time effectively to complete all questions.